

KIVALINA, AK, USA

WMO: 701486

Lat: 67.732N

Lon: 164.548W

Elev: 3

StdP: 101.29

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 26642

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.4	-33.5	-36.5	0.1	-32.3	-34.8	0.1	-31.2	18.1	-17.2	16.4	-16.7	2.1	70	0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	18.5	14.0	16.6	13.0	14.7	12.1	14.8	17.8	13.6	15.9	12.4	14.2	4.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	9.4	15.9	12.3	8.9	14.9	11.2	8.3	13.7	41.4	18.2	38.1	15.8	35.1	14.3	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	14.1	12.1	DB	-38.7	21.8	4.5	2.5	-41.9	23.6	-44.5	25.1	-47.0	26.5	-50.3	28.3	(4)
(5)			WB	-35.6	15.9	2.1	1.3	-37.2	16.9	-38.4	17.7	-39.6	18.4	-41.1	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-17.7	-16.2	-17.4	-9.8	-0.1	6.6	10.2	9.8	5.2	-2.9	-11.7	-15.6	(6)	
	DBStd	12.24	9.37	9.32	6.82	6.57	4.89	4.25	3.16	2.91	3.63	5.18	6.61	7.62	(7)	
	HDD10.0	5534	857	734	848	594	316	120	36	39	148	399	651	792	(8)	
	HDD18.3	8483	1116	967	1107	844	573	353	254	266	394	658	901	1051	(9)	
	CDD10.0	95	0	0	0	0	1	18	41	32	3	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0	(11)	
	CDH23.3	11	0	0	0	0	0	9	2	0	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	3	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.5	5.9	6.0	5.6	5.2	4.6	4.5	4.8	5.5	5.7	5.8	6.1	6.7	(14)	
(15) Precipitation	PrecAvg	452	32	35	26	21	17	20	49	72	50	60	39	31	(15)	
	PrecMax	630	68	111	126	55	46	37	98	155	111	116	116	78	(16)	
	PrecMin	324	7	5	6	3	1	9	12	14	13	24	6	10	(17)	
	PrecStd	79	16	26	24	15	12	8	20	33	21	24	25	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	4.3	-0.9	3.7	14.8	22.2	22.2	20.6	15.1	9.3	2.2	1.3	(19)	
		MCWB	0.7	0.8	-1.6	1.8	9.6	14.1	16.1	15.1	10.3	4.6	1.1	0.2	(20)	
	2%	DB	0.2	1.3	-2.2	2.4	10.3	17.3	18.6	17.8	12.8	6.3	0.8	0.0	(21)	
		MCWB	-0.8	-0.5	-3.0	1.1	7.0	11.8	14.6	14.1	9.7	4.5	0.2	-0.7	(22)	
	5%	DB	-1.7	-0.8	-3.6	1.5	7.9	14.6	16.6	16.1	11.4	4.9	-0.4	-1.5	(23)	
		MCWB	-2.5	-1.5	-4.5	0.4	5.3	11.2	13.5	13.1	9.5	3.7	-1.0	-2.2	(24)	
	10%	DB	-3.7	-2.6	-6.7	0.4	5.5	12.5	14.7	14.1	10.4	3.6	-1.8	-3.8	(25)	
		MCWB	-4.7	-3.5	-7.5	-0.5	3.7	10.1	12.5	11.9	8.8	2.4	-2.4	-4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.4	-1.3	1.9	9.6	15.1	16.7	16.2	11.7	6.0	1.4	0.7	(27)	
		MCDB	2.1	3.2	-1.1	3.2	13.7	20.0	21.1	19.0	13.4	7.6	1.9	1.0	(28)	
	2%	WB	-0.7	-0.2	-2.8	1.2	7.6	12.8	15.0	14.7	10.7	4.8	0.2	-0.8	(29)	
		MCDB	0.0	1.2	-2.3	2.2	10.3	15.9	18.2	17.3	12.1	6.0	0.7	-0.3	(30)	
	5%	WB	-2.3	-1.5	-4.4	0.6	5.4	11.4	13.9	13.5	9.9	3.8	-1.0	-2.3	(31)	
		MCDB	-1.6	-0.6	-3.5	1.4	7.7	14.2	16.3	15.5	11.3	5.0	-0.6	-1.7	(32)	
	10%	WB	-4.3	-3.4	-7.2	-0.6	3.8	10.2	12.7	12.2	8.9	2.5	-2.4	-4.5	(33)	
		MCDB	-3.5	-2.4	-6.5	0.2	5.5	12.8	14.5	13.8	10.3	3.5	-1.8	-3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.1	8.4	8.4	5.9	5.9	5.2	5.5	5.5	5.0	5.7	6.6	(35)	
		MCDBR	7.4	7.9	7.9	6.9	8.8	9.9	8.8	8.3	6.3	4.3	4.0	5.6	(36)	
	5% WB	MCWBR	6.9	7.0	7.7	6.0	6.1	6.0	5.5	5.4	4.3	3.5	4.0	5.4	(37)	
		MCWBR	7.2	7.8	7.7	6.6	8.3	9.1	7.8	7.2	5.4	3.9	3.8	5.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.171	0.235	0.258	0.322	0.361	0.346	0.355	0.363	0.306	0.253	N/A	N/A	(40)	
		taud	1.899	2.078	2.213	2.056	2.075	2.391	2.452	2.412	2.519	2.282	N/A	N/A	(41)	
	Ebn at Noon		146	579	797	813	819	856	836	781	754	566	N/A	0	(42)	
		Edh at Noon	10	48	76	120	132	100	92	86	60	39	N/A	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.05	0.55	2.28	4.48	5.76	6.15	4.78	3.44	2.06	0.68	0.10	0.00	(44)	
	RadStd		0.01	0.07	0.17	0.23	0.49	0.52	0.50	0.43	0.21	0.09	0.01	0.00	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (8 neighbors)		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page